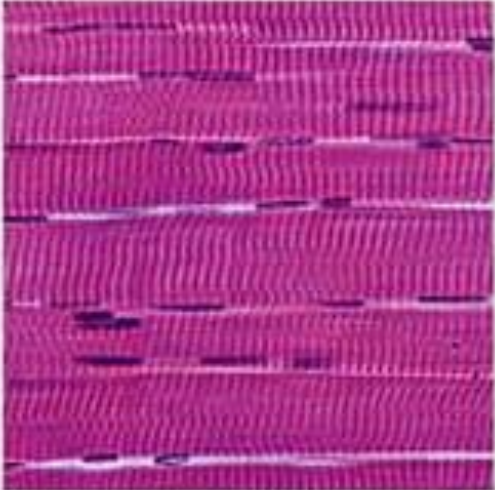
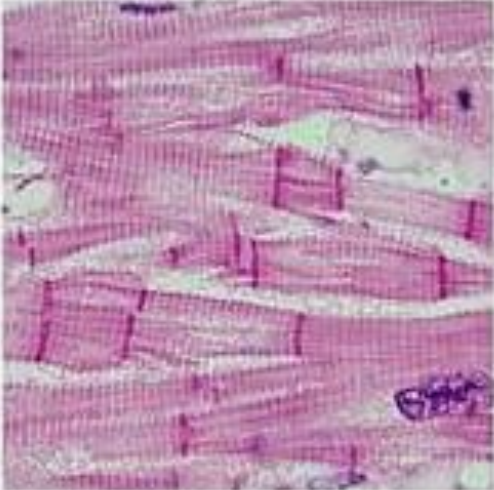




Skeletal, Cardiac and Smooth Muscle Histology

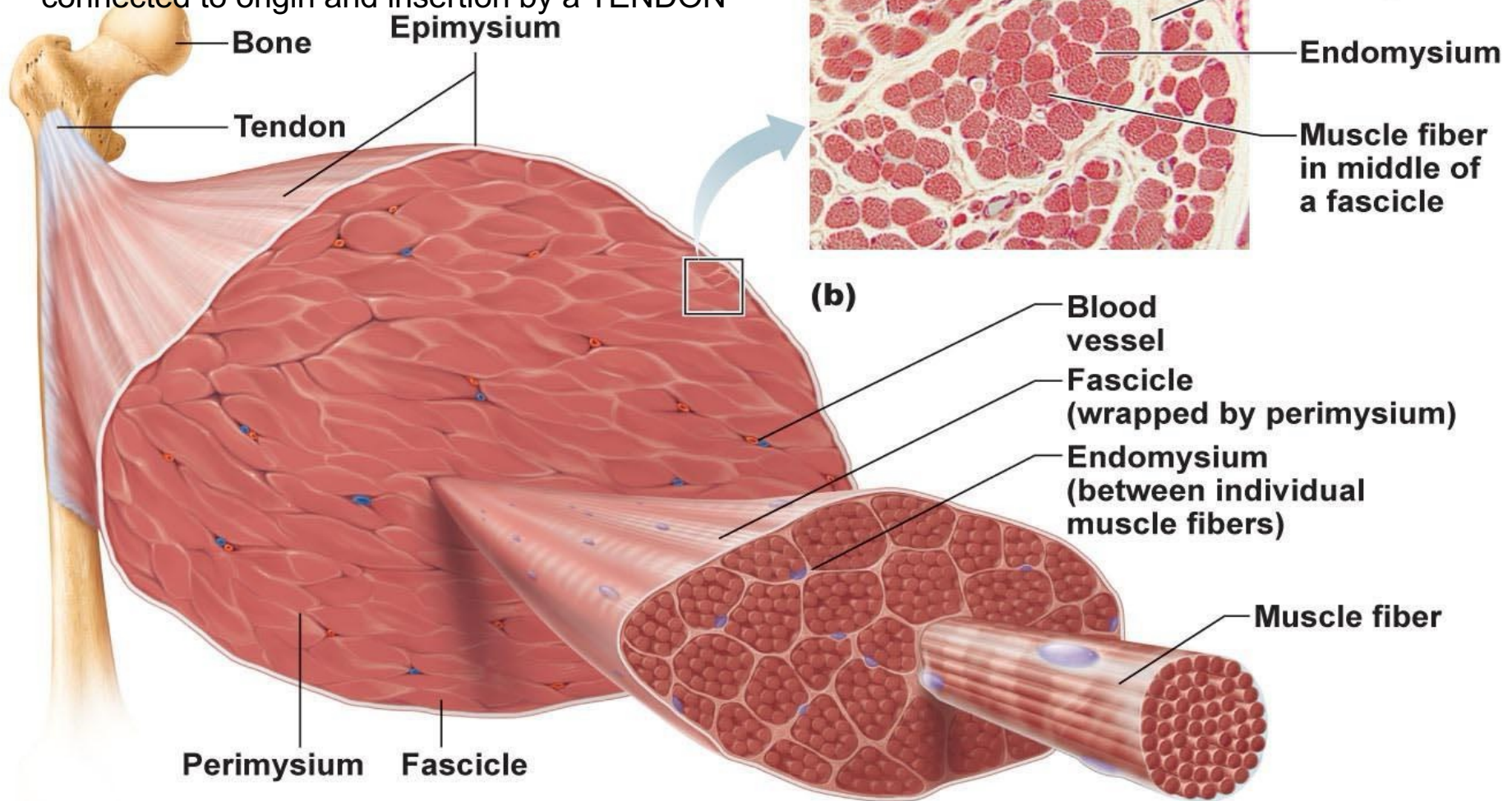
Thomas F. Freddo, O.D., Ph.D.
Professor
MCP Health Sciences University

Three Main Categories – 2 groups

Characteristic	Skeletal	Cardiac	Smooth
	STRIATED		
Body location	Attached to bones or, for some facial muscles, to skin	Walls of the heart	Mostly in walls of hollow visceral organs (other than the heart)
			
Cell shape and appearance	Single, very long, cylindrical, multinucleate cells with very obvious striations	Branching chains of cells; uninucleate, striations; intercalated discs	Single, fusiform, uninucleate; no striations
	 	 	 

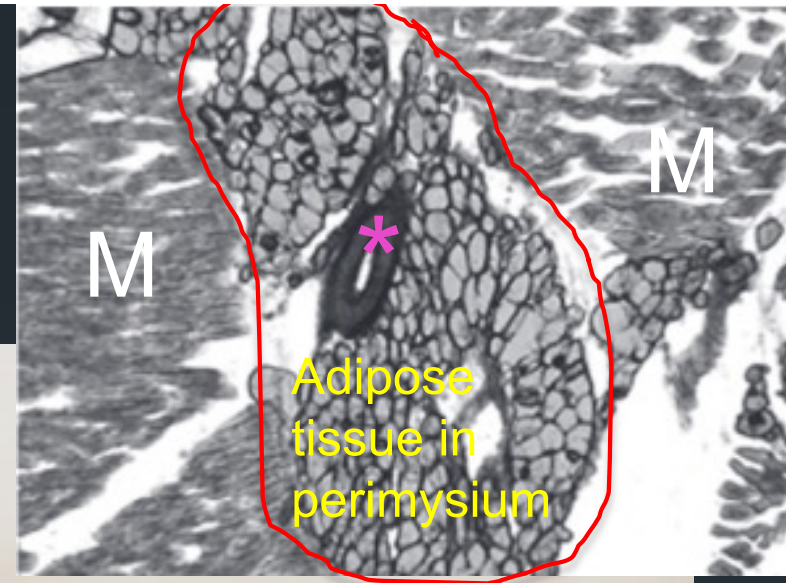
Skeletal Muscle Cells

- Individual muscle fibers surrounded by **ENDOMYSIUM**
- Fibers are grouped together into fascicles and fascicles are surrounded by **PERIMYSIUM**
- The whole muscle is wrapped in **EPIMYSIUM** and connected to origin and insertion by a **TENDON**



(a) Perimysium provides continuous passageways for nerve branches and small vessels to reach the interior of the muscle. Occasionally, the perimysium may contain clusters of adipose tissue.

Wagyu Beef

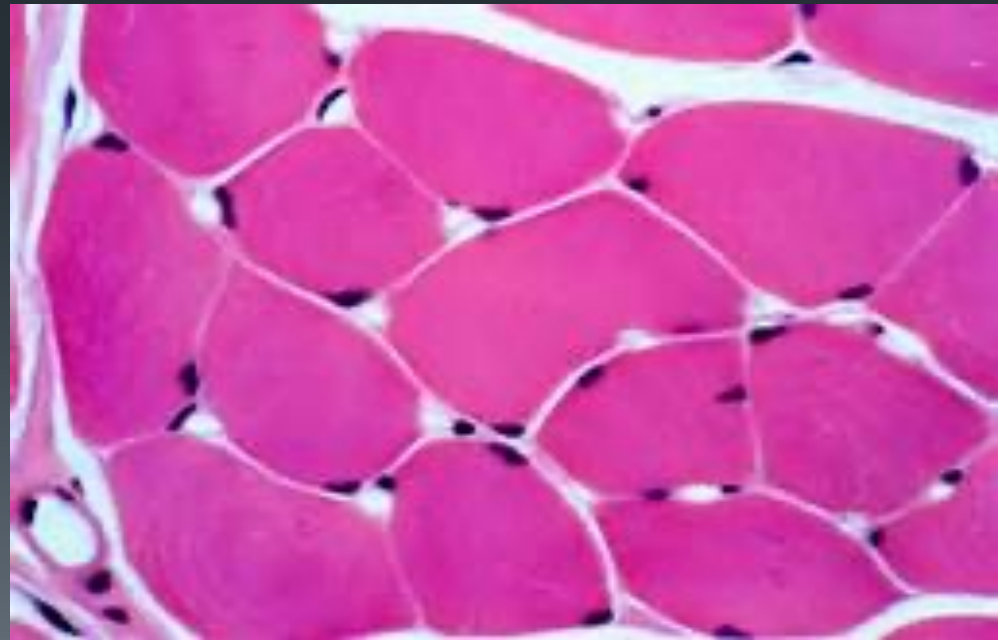
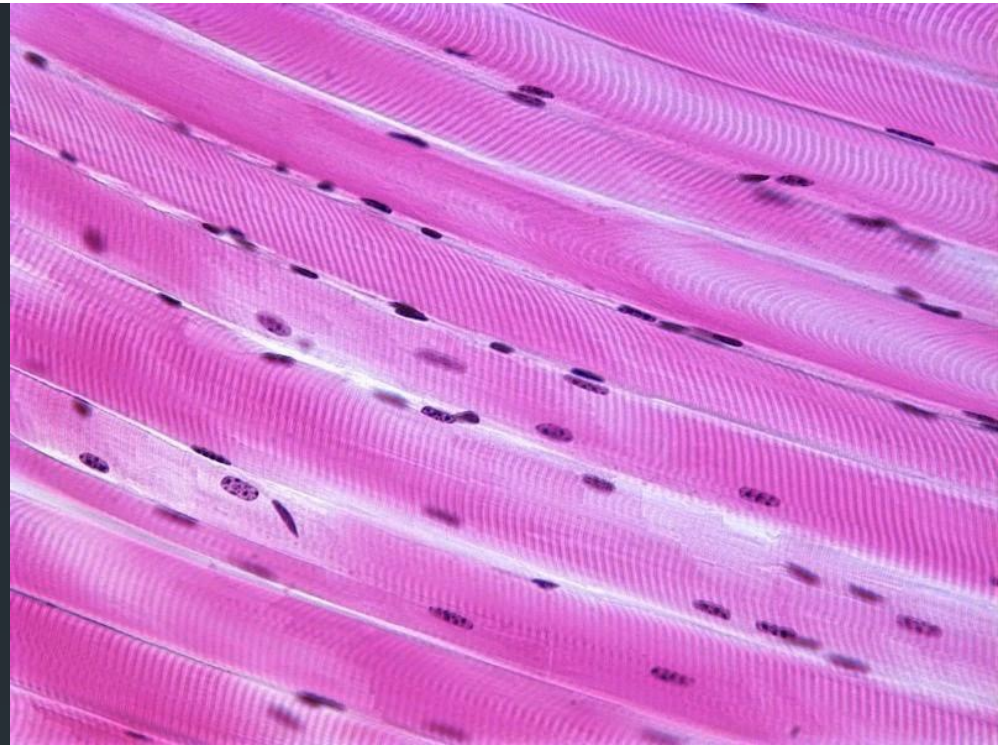


Cattle are bred to develop lots of adipose tissue within their perimysium. (aka - marbling)

Skeletal Muscle Cells

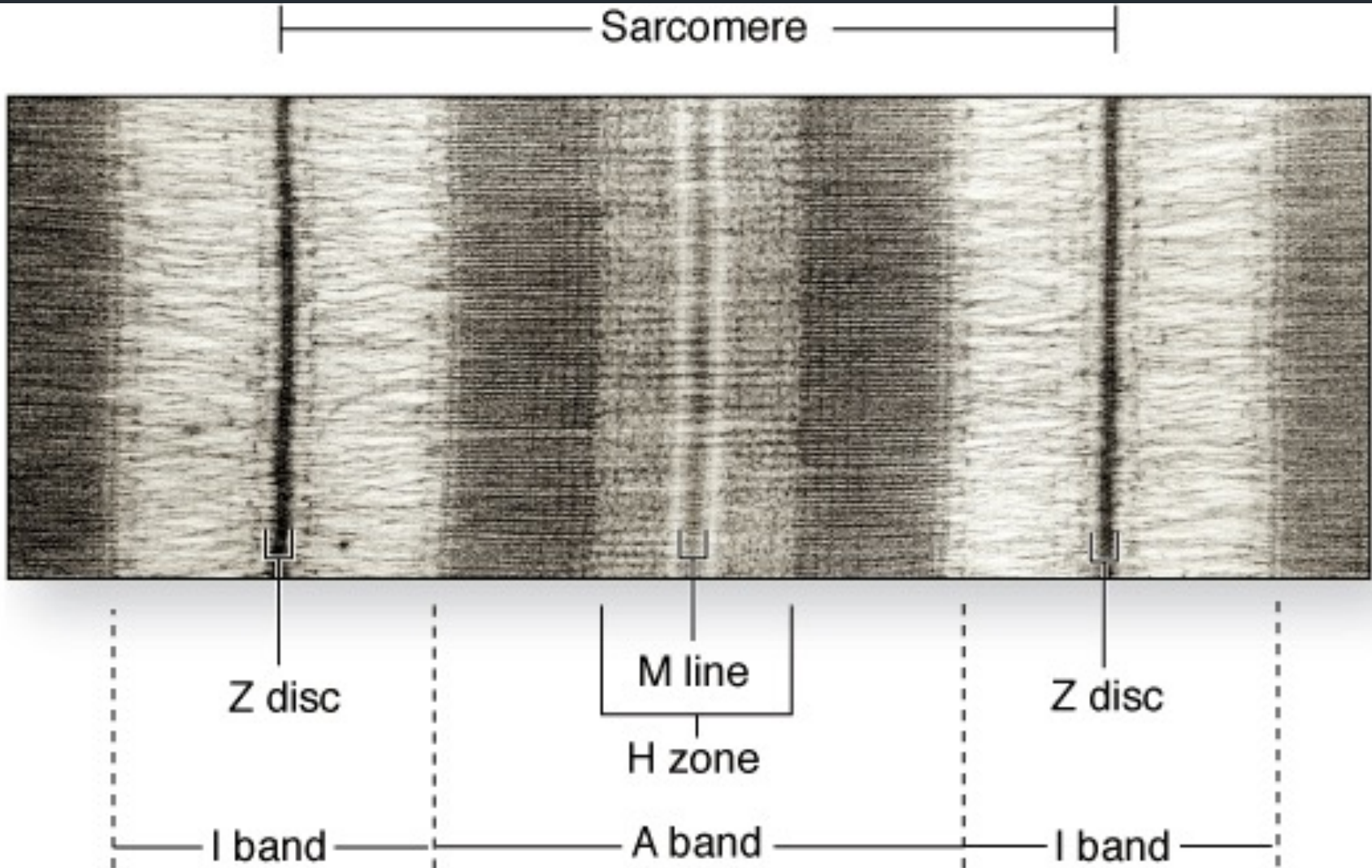
Light microscopy

- The functional units within these cells are termed sarcomeres.
- In cross-section its clear that skeletal muscle cells are multinucleate.
- In skeletal muscle, as more force is required, more cells are recruited to contract.
- By contrast, in cardiac muscle EVERY cell contracts with each beat. If more force is required, each cell contracts harder until it reaches its limit. Harder contraction of cardiac muscle is called inotropy.



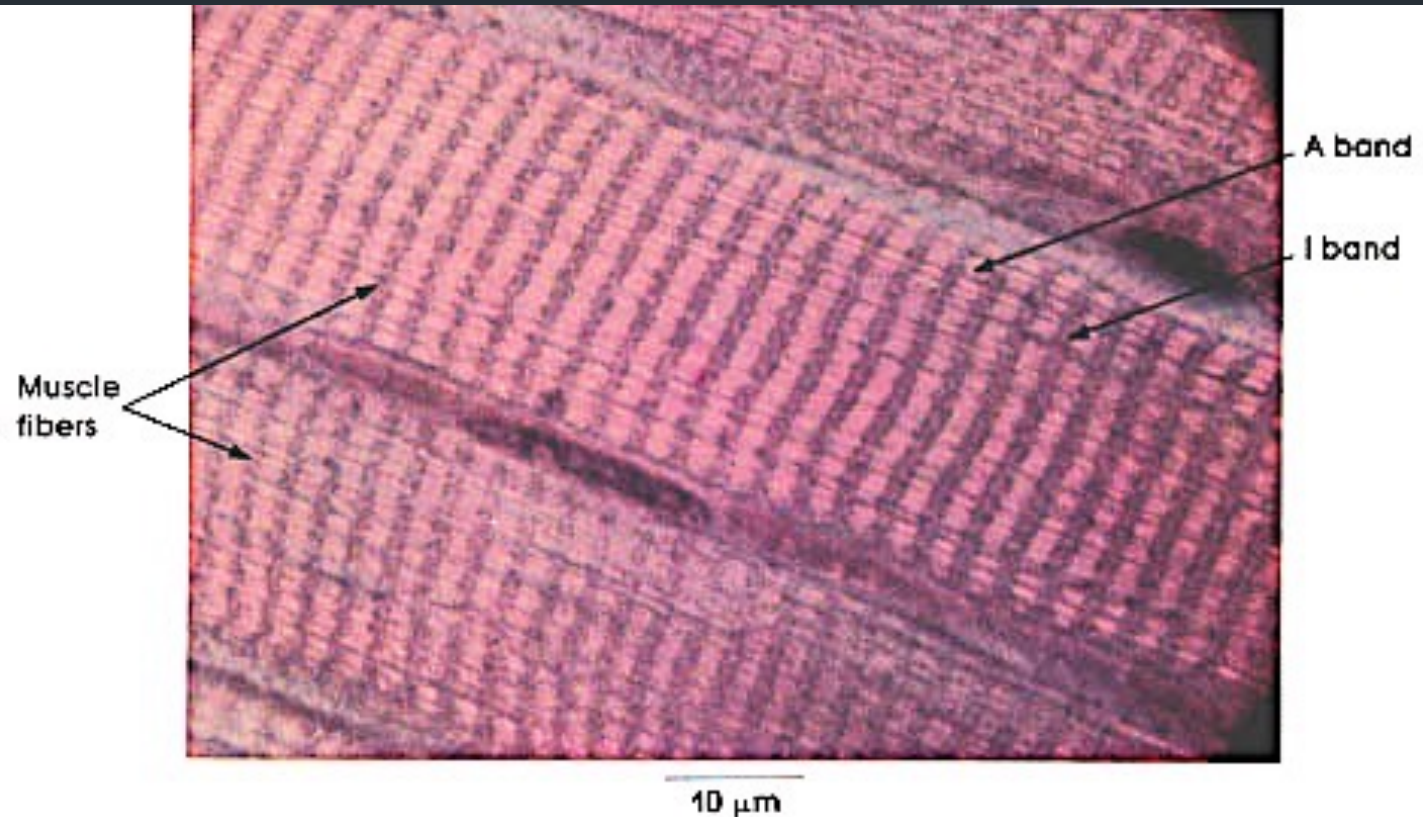
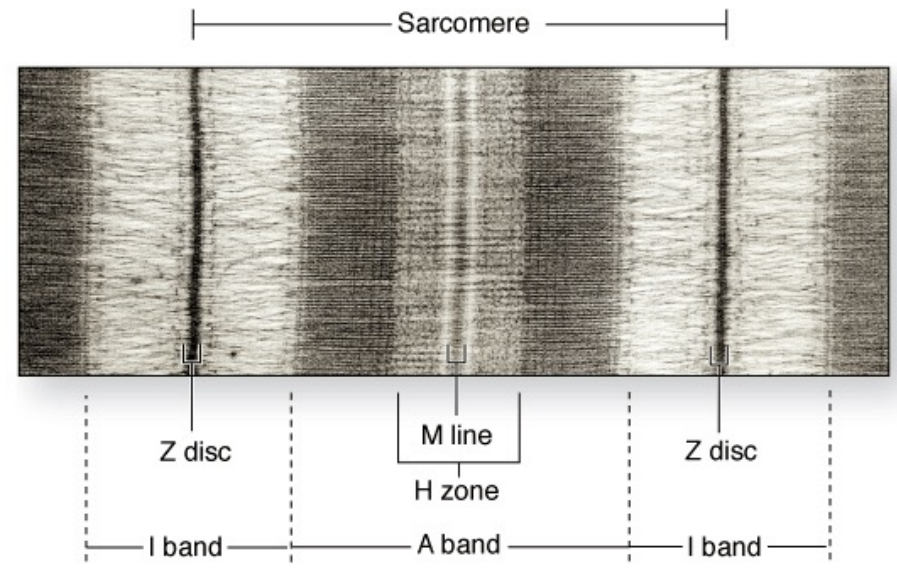
Skeletal Muscle Cells – electron microscopy

- A sarcomere is defined as the contractile unit between adjacent Z-discs



Nomenclature – mostly from German

- M – mittle = middle band
- H – helle = bright
- Z – zwischen = between
- A – anisotropic (light appearance under polarized light)
- I – isotropic (dark)



Cross-section of sarcomere

Sarcomeres are composed of:

Thick fibers –

Myosin

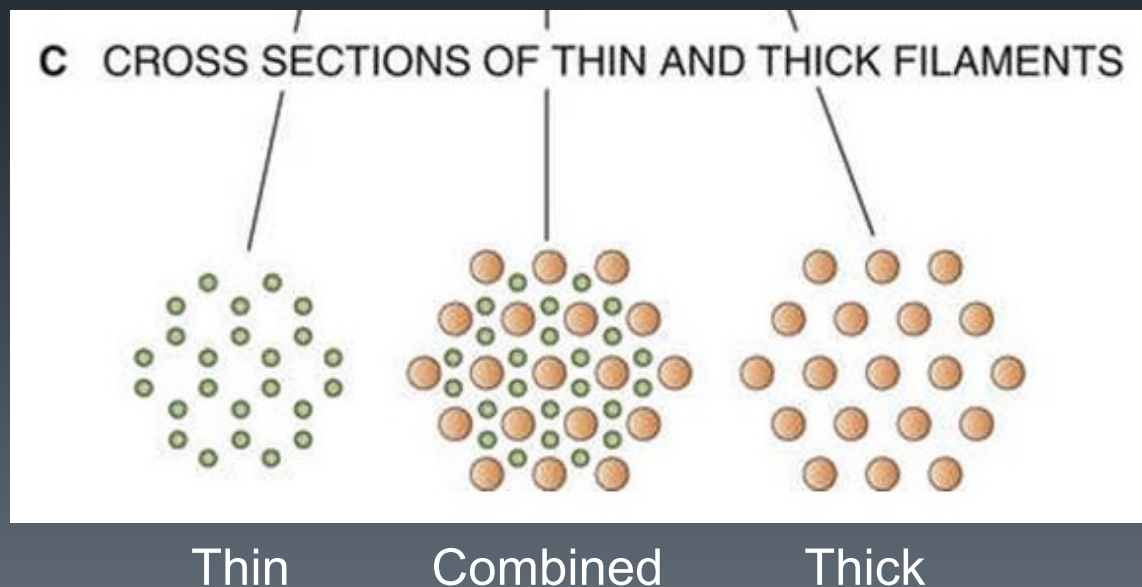
Thin fibers –

Actin – globular protein with myosin-binding site

Tropomyosin – runs along actin strands to block myosin-binding site

Troponin – globular protein which binds Ca^{++} .

Once bound, a conformational change occurs that removes tropomyosin inhibition of actin myosin interaction.



Longitudinal Section of the Sarcomere

Sarcomere –
Z-disc to Z-disc (one A-band and two halves of I-bands)

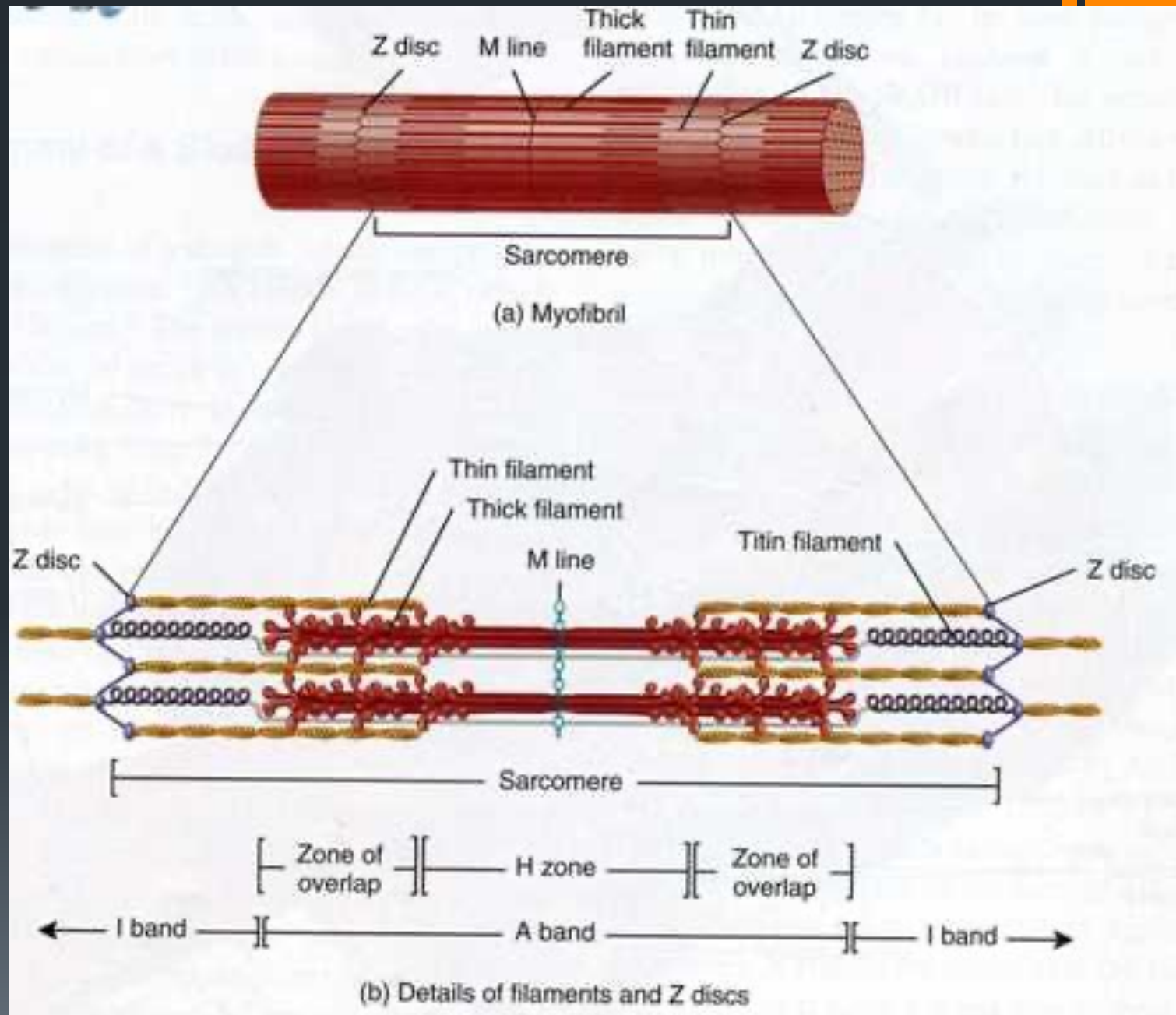
A band – length of thick filaments in sarcomere

I band – portion of sarcomere with no myosin (between A bands)-straddles Z-disc

M-line – unmoving center of sarcomere links ends of thick filaments

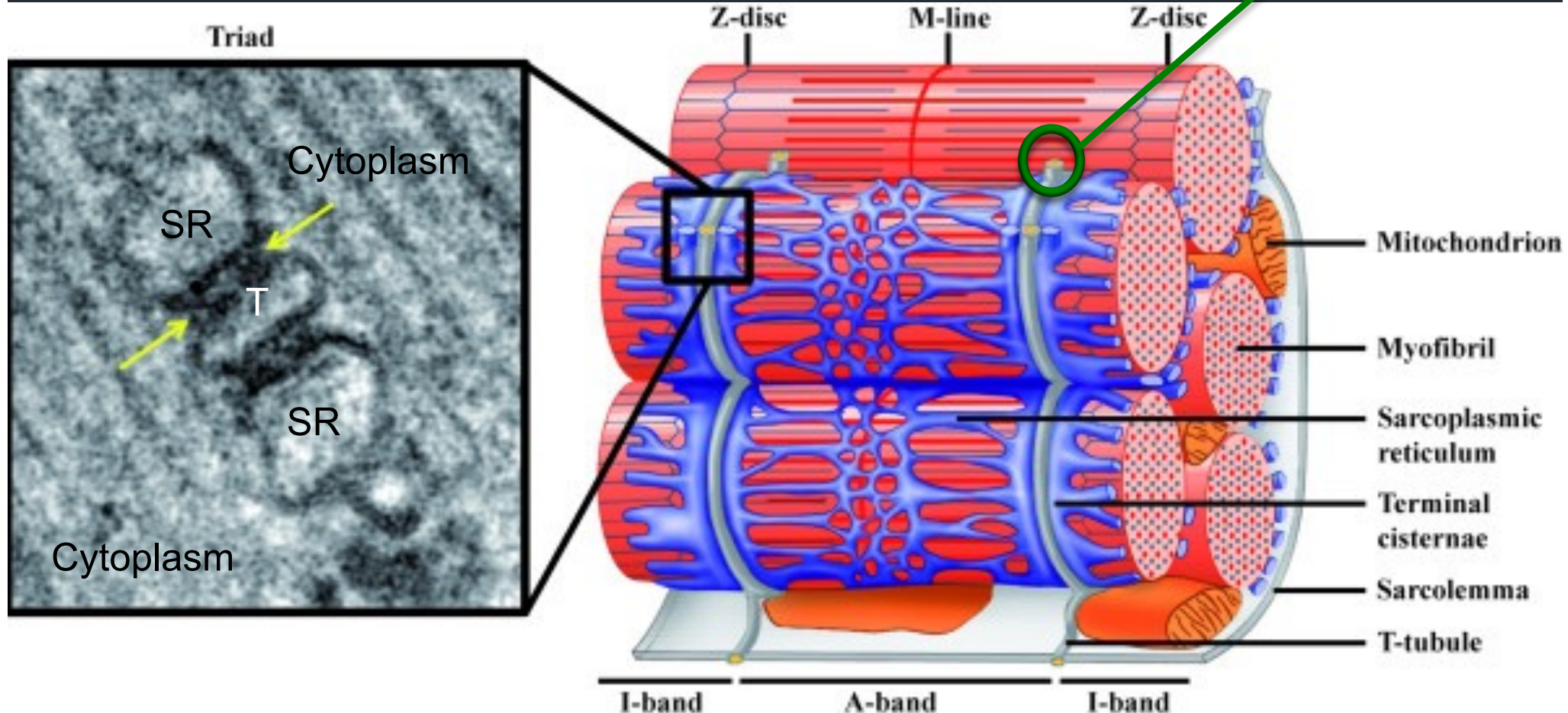
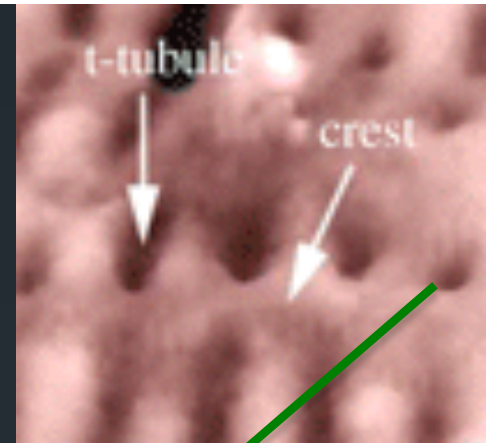
H-zone – length of non-overlap of thick and thin filaments

oooo – Titin coils fixing thick filaments to Z-disc
A-actinin (not shown) serves same function for thin filaments



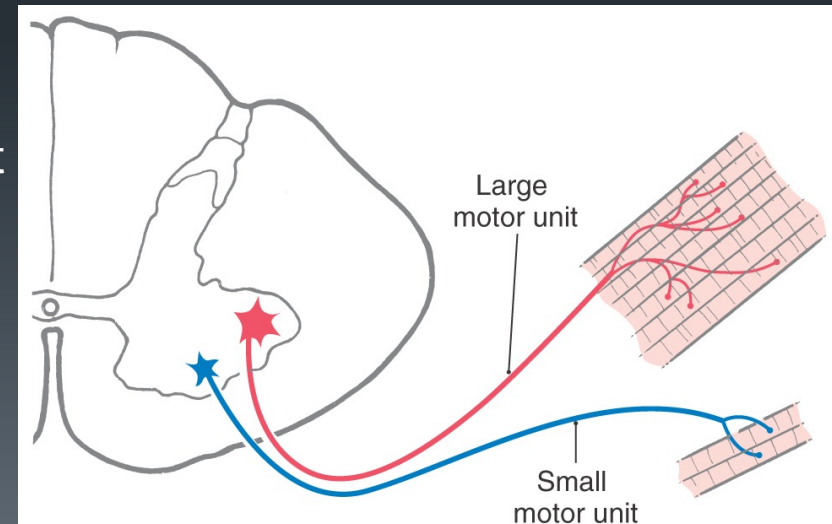
T-Tubule system and Sarcoplasmic Reticulum

- Skeletal myocytes have T-tubules, that surround the cell at the junction of A and I bands. They are invaginations of cell membrane that dive into the core to carry the action potential of the cell membrane internally to contact the terminal cisternae of *two* adjacent lengths of sarcoplasmic reticulum (called a Triad). The sarcoplasmic reticulum is a system of channels that, when stimulated by an action potential, release pools of calcium into the cytoplasm for contraction and then reabsorb the Ca^{++} for relaxation.



Extraocular Muscles

- Extraocular muscles differ histologically from most other skeletal muscles in that they are made up of 2 (and possibly 3) different types of muscle cells. Both contain sarcomeres.
 - Fast-twitch** muscle fibrils generate fast eye movements and are composed of well-defined myofibrils with well-developed sarcomeres.
 - Slow-twitch** muscle fibrils generate slow or tonic eye movements and are composed of poorly defined myofibrils with poorly developed sarcomeres.
- Cholinergic motor neurons supply both types of muscle fibers.
 - The innervation to Fast-twitch fibrils is thick and heavily myelinated, with a single neuromuscular junction.
 - The innervation to slow-twitch fibrils is thin, with multiple grapelike clusters of neuromuscular junctions.
- EOMs have the smallest motor units of any skeletal muscles. This provides the remarkable precision of contraction and coupled relaxation of the muscle that pulls in the opposite direction (yoked).



Fast twitch vs Slow twitch

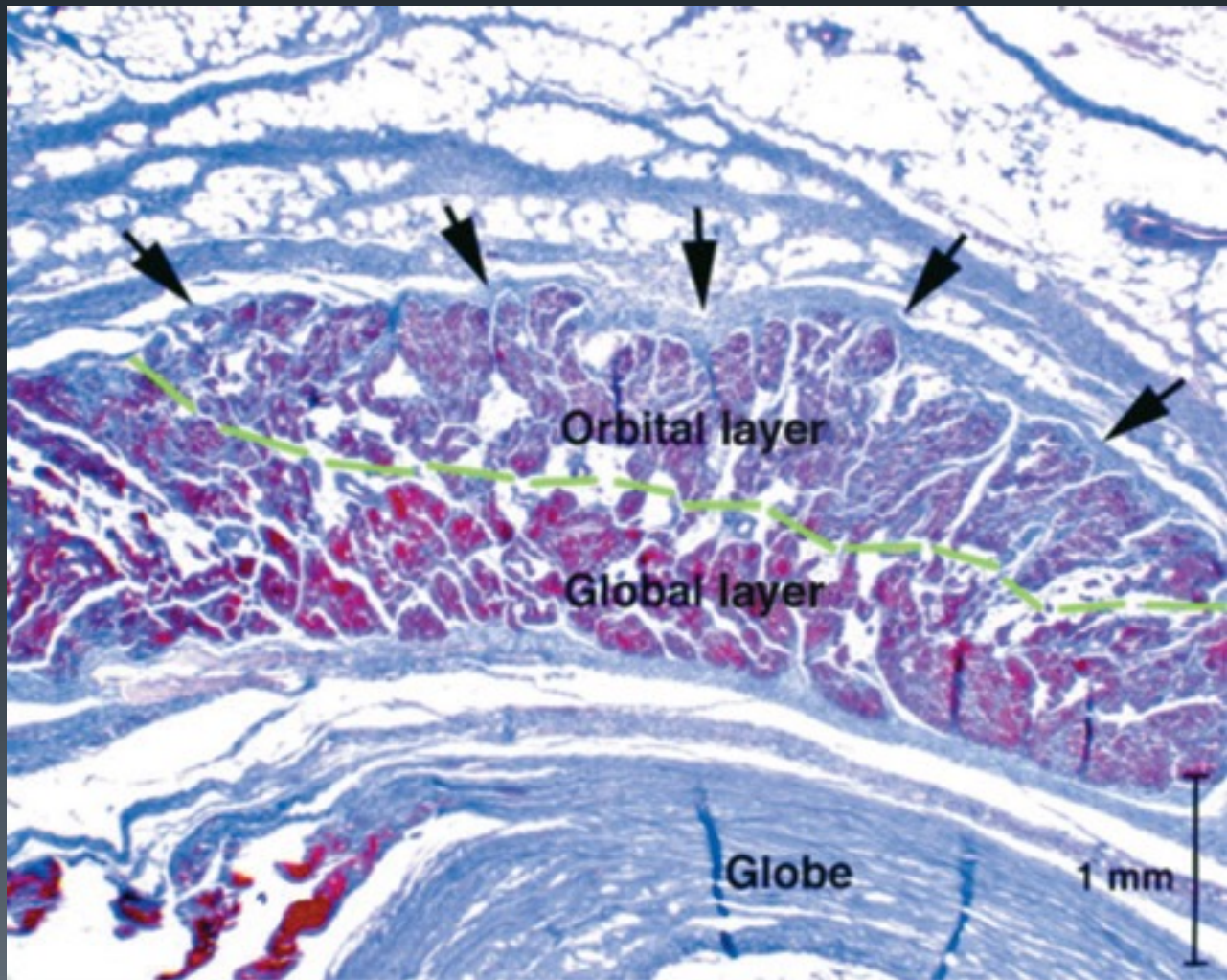
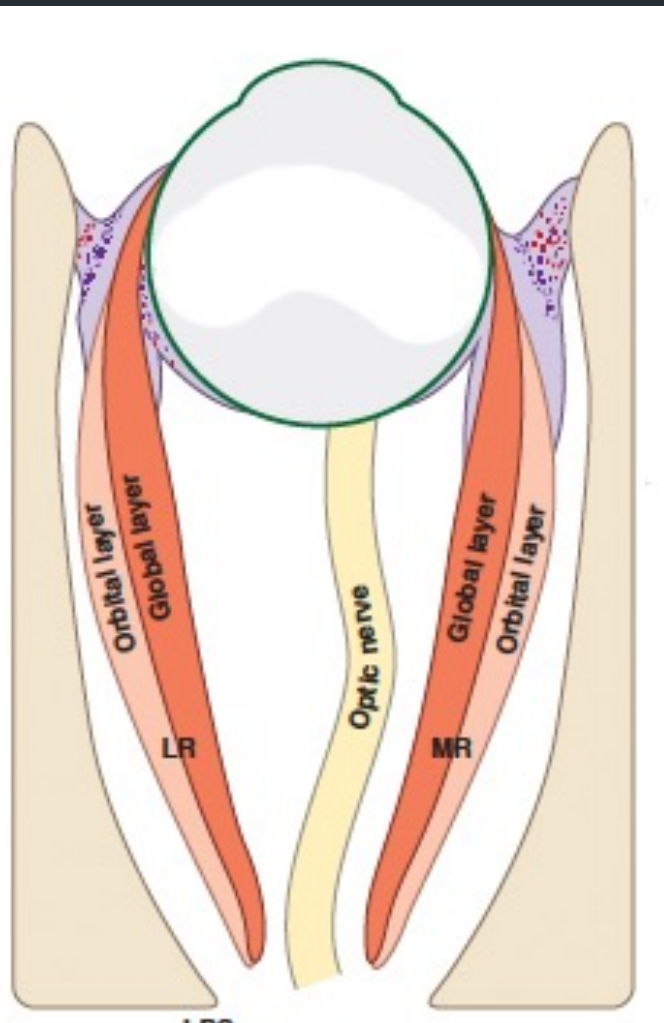
- To experience the difference between fast twitch ("white", not so much myoglobin) and slow twitch ("red", lots of myoglobin), try the following. First, blink your eyelid. Now "completely blink" your anal sphincter. That's the difference between "fast" and "slow".

Extraocular Muscles

Two histologically distinct portions of most rectus muscles are found – GLOBAL LAYER and ORBITAL LAYER.

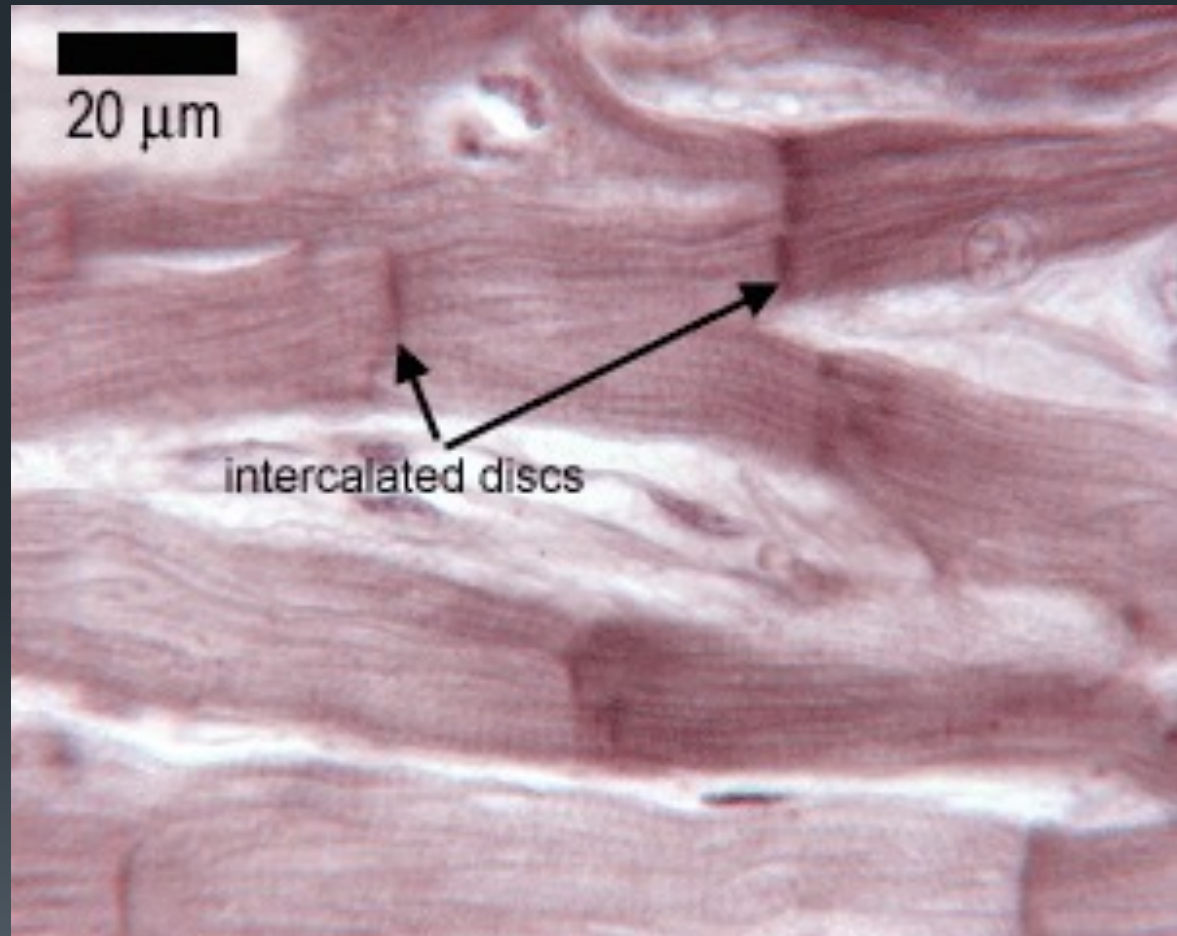
These do not correspond to pure populations of fast and slow twitch fibers. Approximately 62% of the fibers in the global layer and 81% in the orbital layer are FAST fibers.

Most of the rest are SLOW but there may be a third intermediate category that accounts for a small percentage as well.



Cardiac Muscle Cells

- As in skeletal muscle, these cells are composed of sarcomeres but are not multinucleate cells.
- These cells branch but each sarcomere runs from a Z line to a Z line. Z-lines correspond in location to **intercalated discs**.
- REMINDER: In skeletal muscle, as more force is required, more cells are prompted to contract
- In cardiac muscle EVERY cell contracts with each beat. If more force is required, each cell contracts harder until it reaches its limit. Harder contraction is called inotropy.



Cross-section of cardiac sarcomere

Sarcomeres are composed of:

Thick fibers –

Myosin

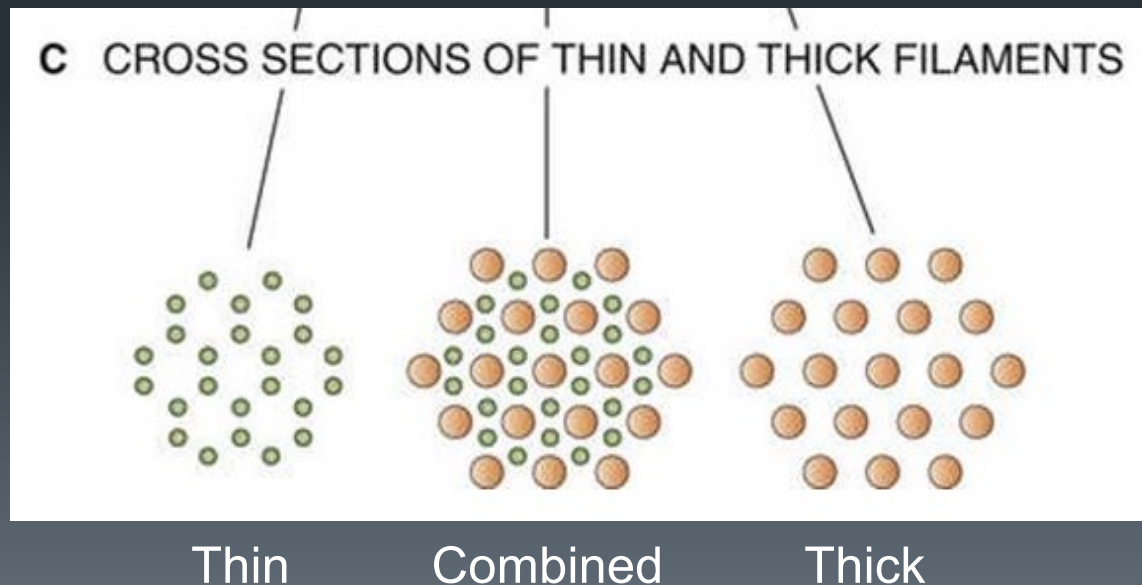
Thin fibers –

Actin – globular protein with myosin-binding site

Tropomyosin – runs along actin strands to block myosin-binding site

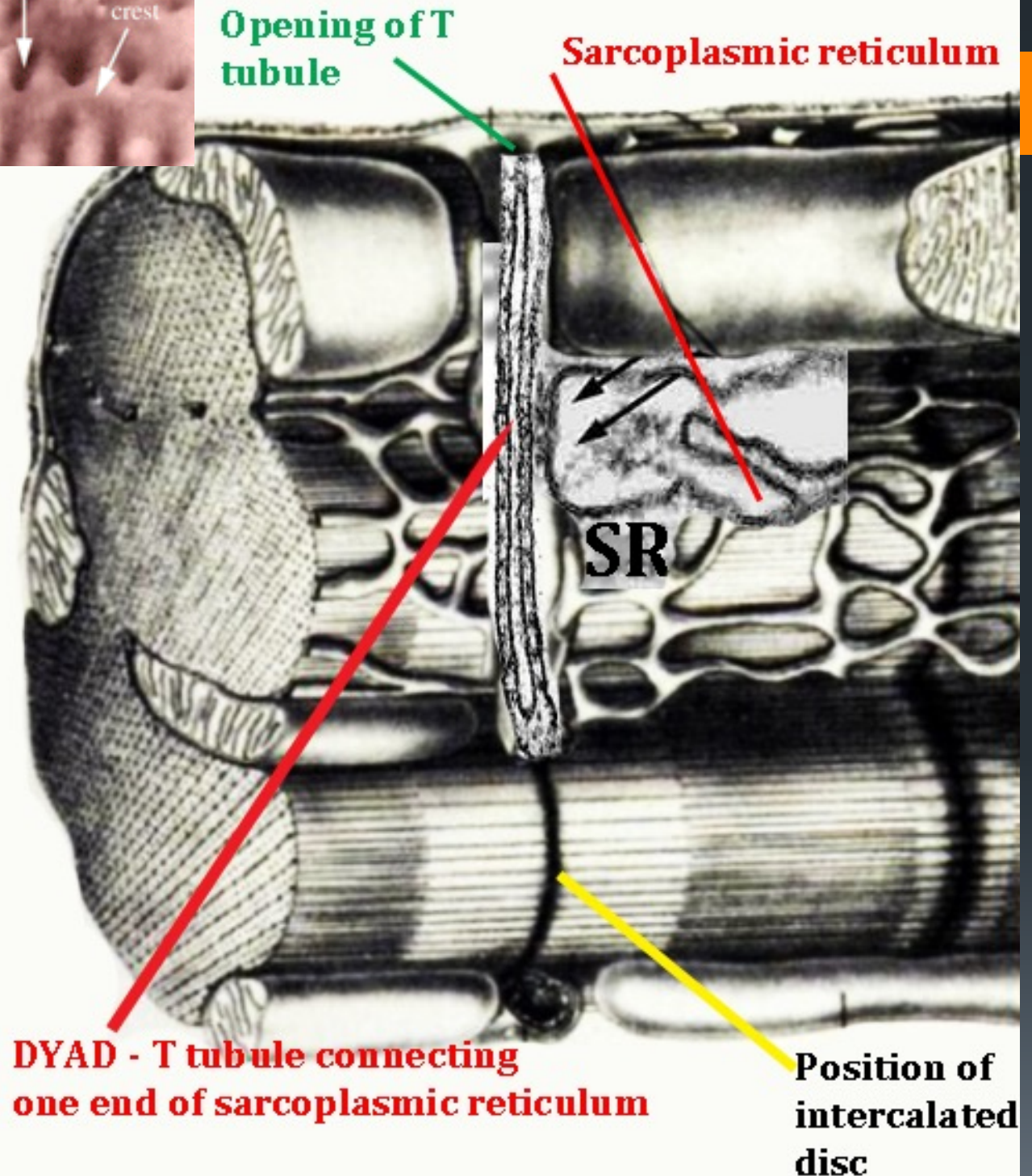
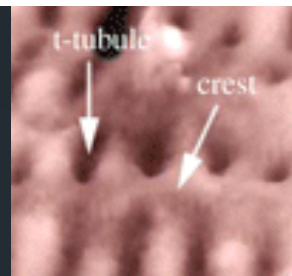
Troponin – globular protein which binds Ca^{++} .

Once bound, a conformational change occurs that removes tropomyosin inhibition of actin myosin interaction.



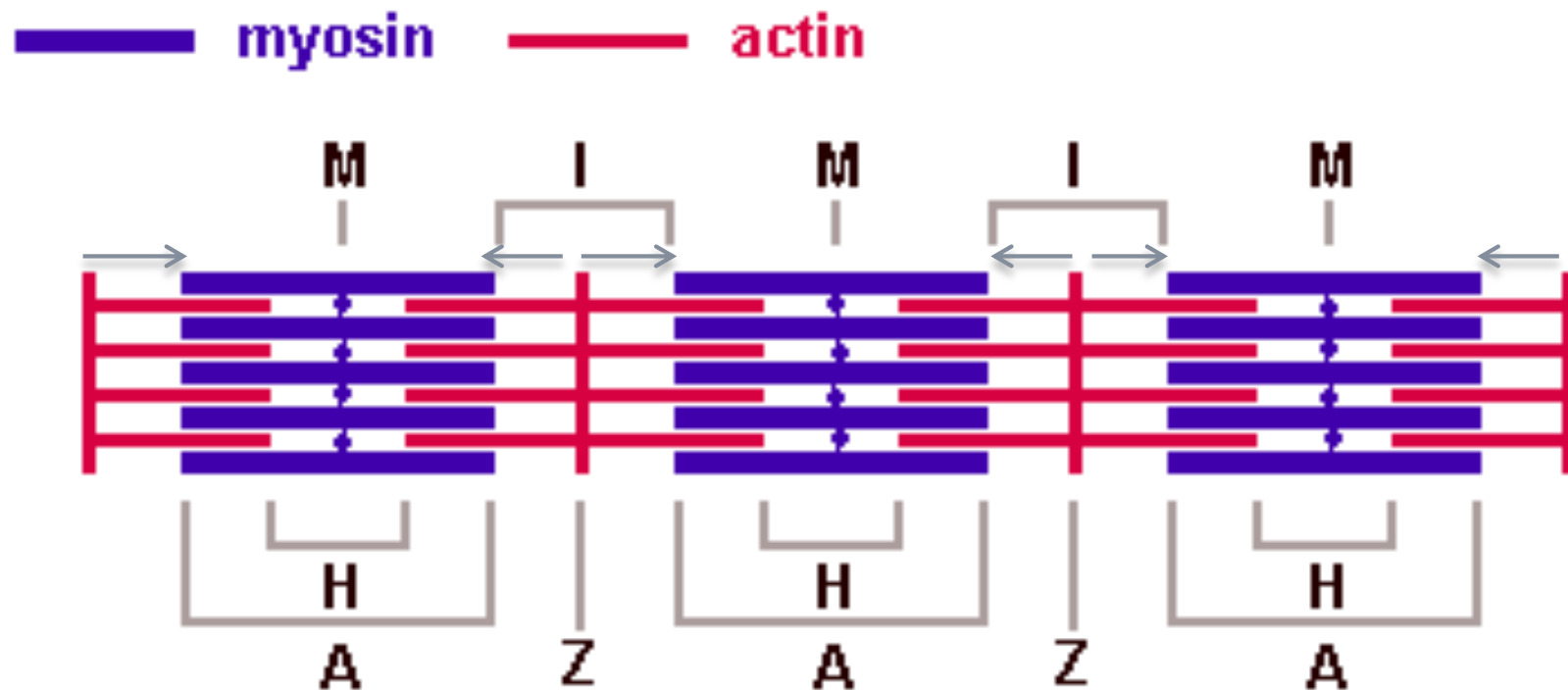
T-Tubule system and Sarcoplasmic reticulum

- Cardiac myocytes have a T-tubules, invaginations of cell membrane that dive deeply to carry the action potential of the cell membrane internally to contact the sarcoplasmic reticulum (called a dyad) for controlled release and reabsorption of Ca^{++} into cytoplasm from the sarcoplasmic reticulum. Tubules of cardiac myocytes are located at the Z lines (Intercalated discs)



SLIDING FILAMENT MODEL

Z lines moves to edges of A band. H and I zones disappear. A zone does not change



Bands and lines in the contractile apparatus of skeletal muscle

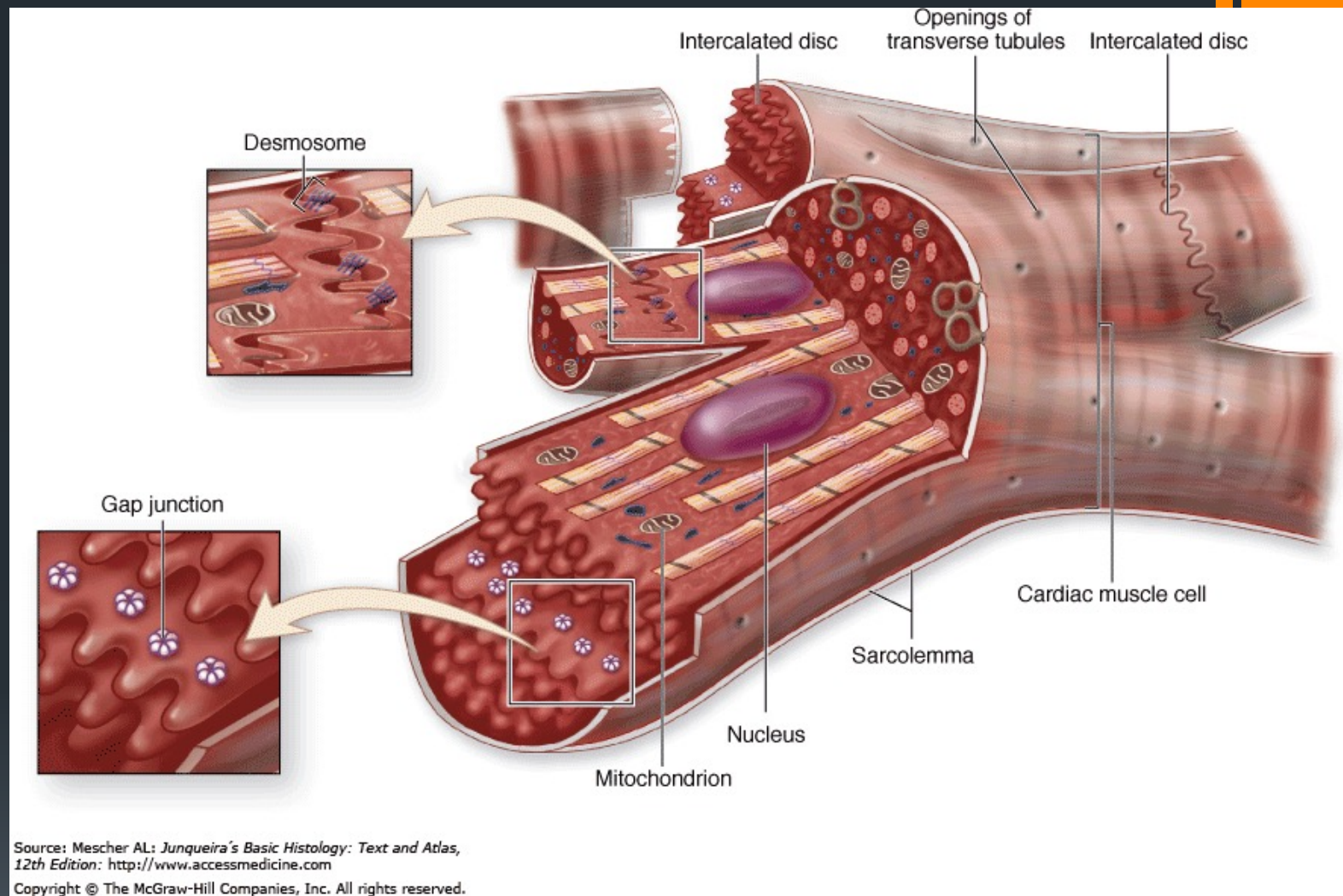
Holding Sarcomeres Together

- Intercalated discs are composed of 3 junctions:

Desmosomes – ensure that sarcomeres remain together during pull of contraction

Gap junctions: allow action potential to spread along a linear row of cardiomyocytes during contraction

Fascia adherens – anchoring sties for actin



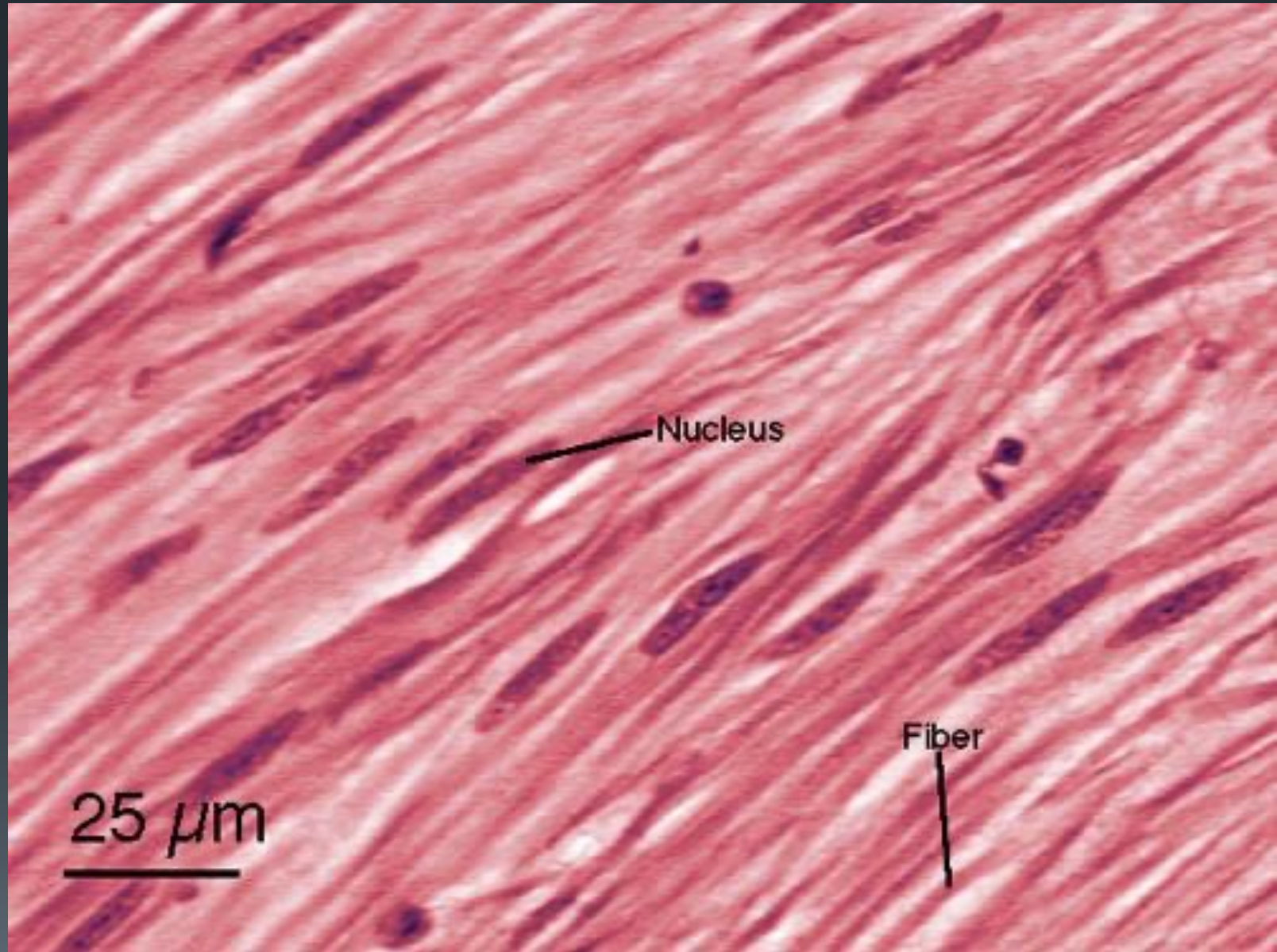
Intercalated discs have very low electrical resistance and action potentials readily pass from one cardiac muscle cell to the next.

Smooth Muscle

- **Unitary Smooth Muscle** – GI tract, bladder, uterus, ureter and iris sphincter
 - Cells are linked by gap junctions
- **Multiunit Smooth Muscle** – ciliary muscle and vas deferens
 - The muscle fibers behave as individual motor units like skeletal muscle fibers and are densely innervated by post-ganglionic fibers of parasympathetic and sympathetic autonomics.
- **Vascular smooth muscle shows features of both**
- Smooth muscle lacks the absolute strength of skeletal muscle but can sustain contraction for longer periods (e.g. the levator palpebrae superioris lifts the lid. Mueller's muscle (smooth) is what largely keeps the lid open between blinks.

Smooth Muscle

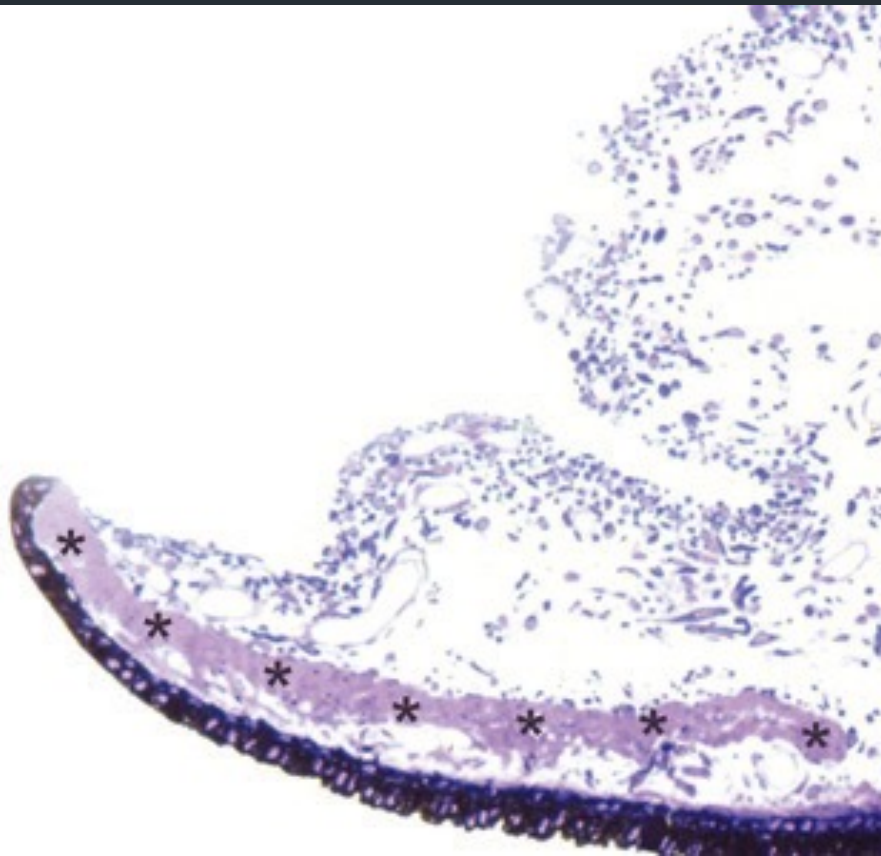
Long fusiform nuclei and no striations across cells.



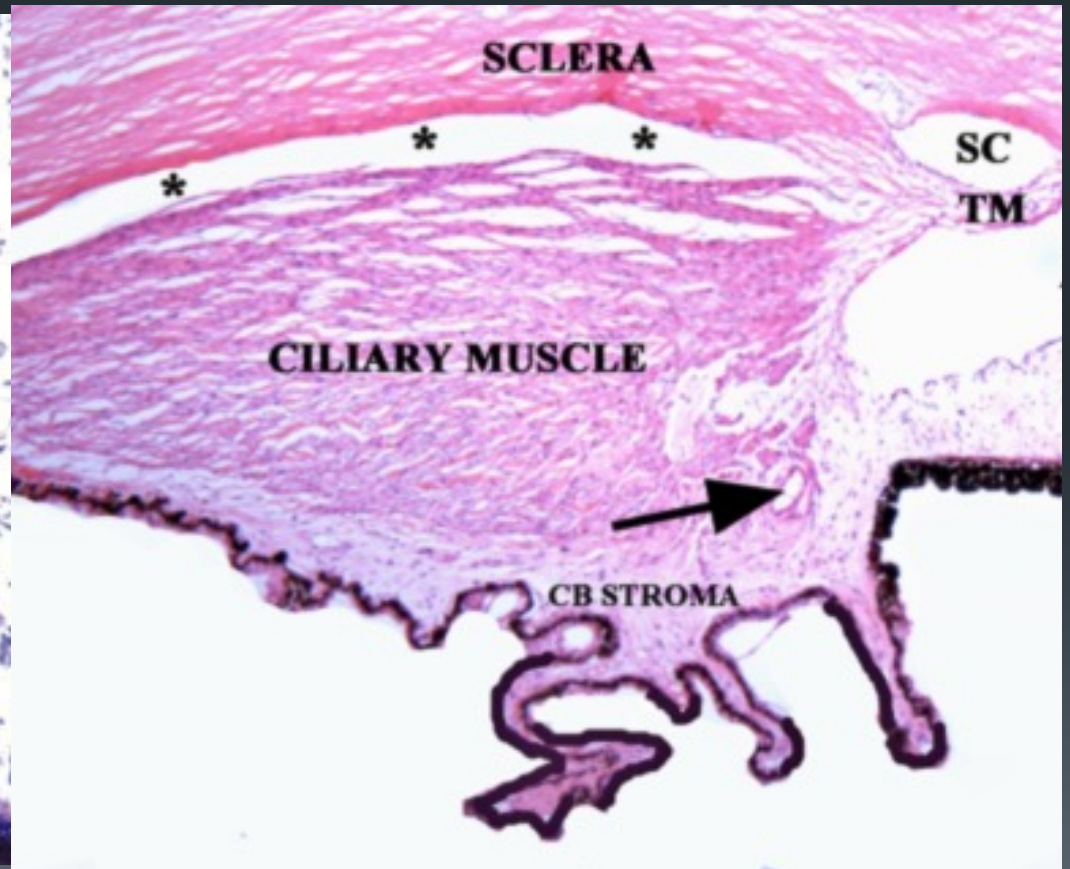
Smooth Muscles in the Eye



Iris Sphincter – unitary
Gap junctions

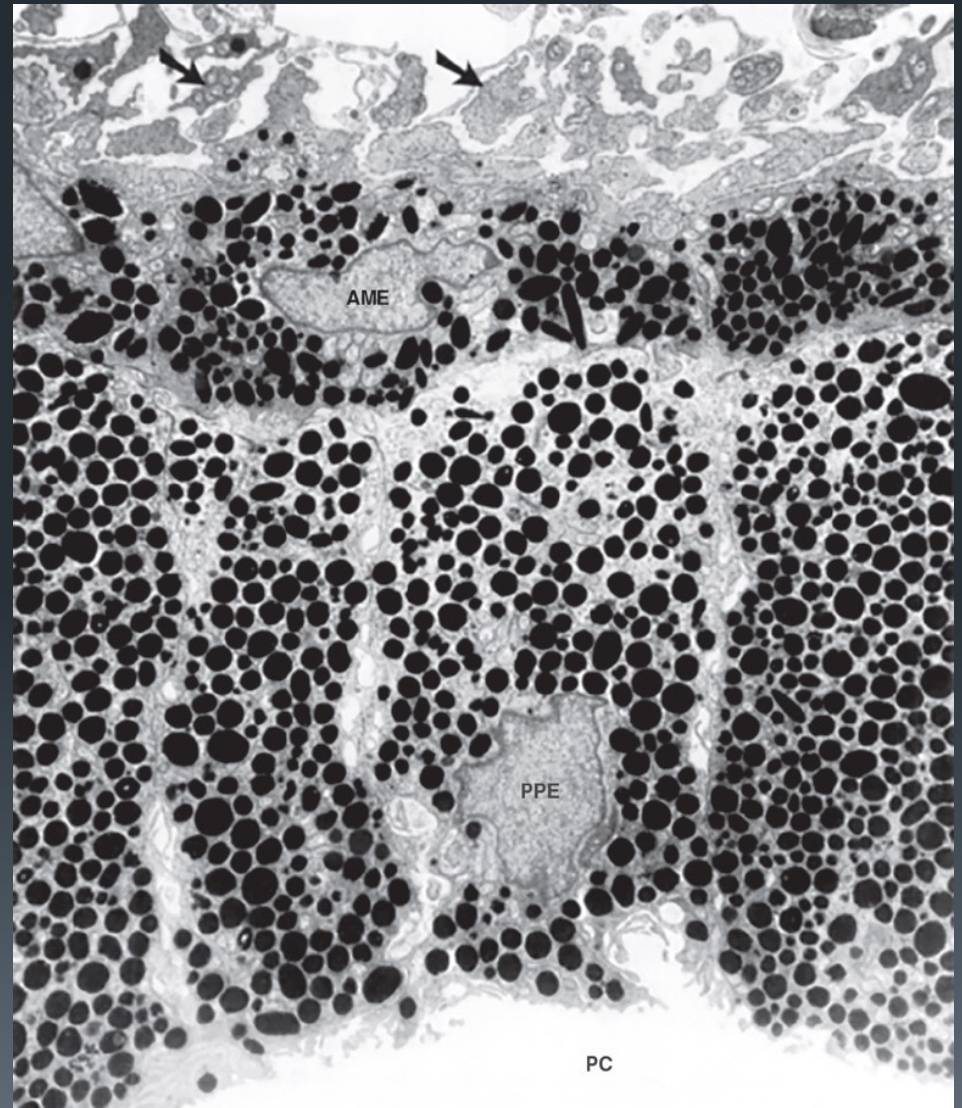
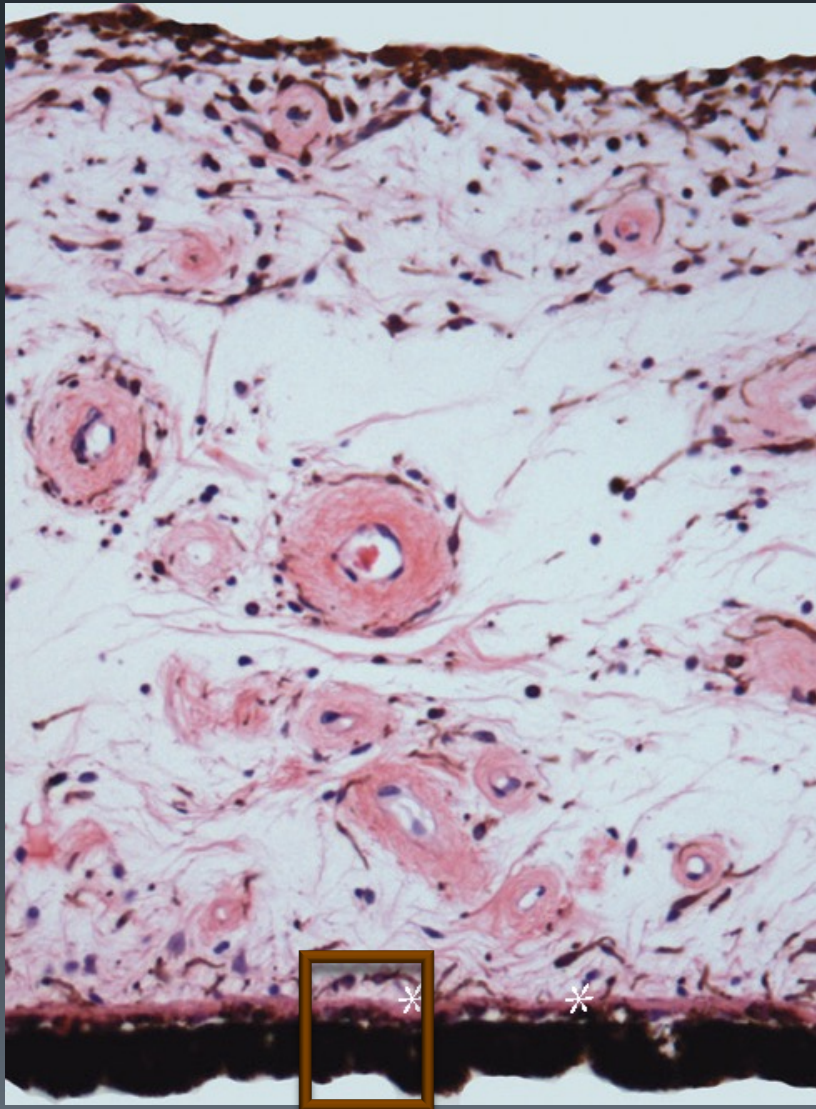


Ciliary muscle – multi-unit
No gap junctions

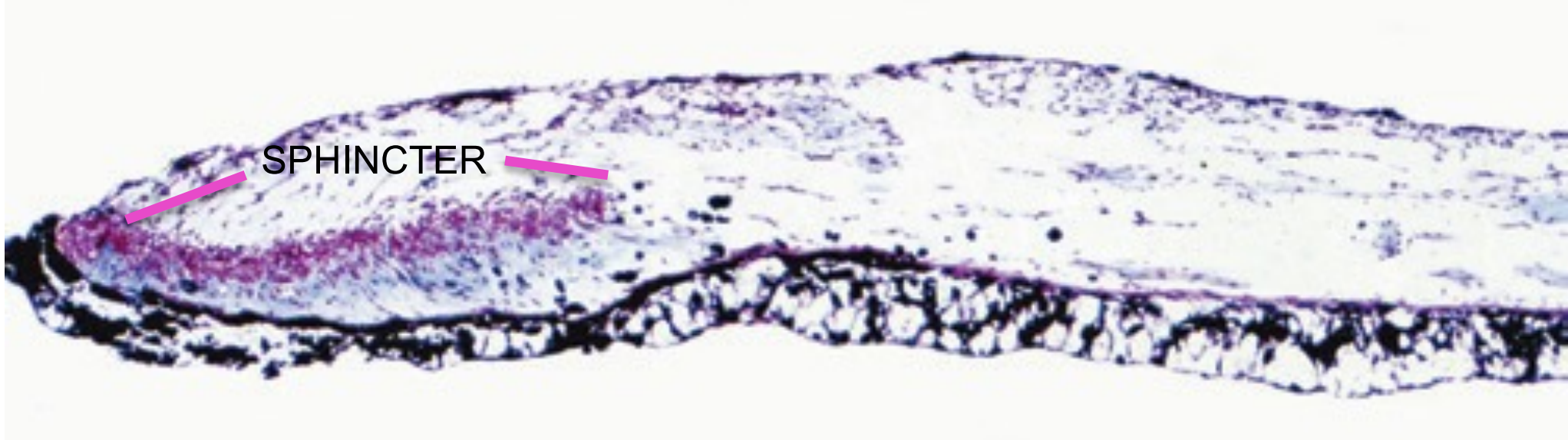


Myoepithelium (usually found in ducts of exocrine glands but also the dilator muscle of the iris)

Myoepithelial cells are true epithelial cells that express keratins but have smooth muscle-like processes that are invariably produced along the basal half of such cells. The myo component expresses α smooth muscle actin and they operate in essentially the same way as smooth muscle, under sympathetic control.



Lacy Vacuolization



It is important to distinguish between myoepithelial tissues and true smooth muscle. In diabetes, the myoepithelial cells accumulate excessive glycogen but true smooth muscle cells (like the sphincter) are not affected.

These deposits compromise cell function, including the “myo” component.

This is part of the reason why diabetics generally don't respond as well to dilating drops but constrict just fine.

Myofibroblasts are found sub-epithelially in many mucosal surfaces. In many organs like liver, lung, and kidney they are primarily involved in fibrosis. Pericytes, such as those found within the tunica media of retinal capillaries, are myofibroblast-like.

Myofibroblasts also play a role in fibrosis and usually stain for the intermediate filament vimentin which is a general mesenchymal marker and "alpha smooth muscle actin" (human gene = *ACTA2*) and for palladin, which is a cytoskeletal actin scaffold protein.

They are likely the source of wound contraction

Myofibroblasts

